

Theory: Autoencoder Neural Network

An **Autoencoder** is an **unsupervised neural network** model used to learn compressed, efficient representations of data. Its main goal is to **reconstruct its input** as accurately as possible after encoding it into a lower-dimensional latent space.

1. Structure and Working

An autoencoder consists of three main parts:

1. **Encoder:** Compresses input data (X) into a smaller latent representation (Z).

$$Z = f(W_1X + b_1)$$

where (f) is a nonlinear activation (like ReLU or sigmoid).

2. **Latent Space (Code):** The compressed intermediate feature space that captures essential data patterns.
3. **Decoder:** Reconstructs the original input from the latent code.

$$\hat{X} = g(W_2Z + b_2)$$

The output ($\hat{X}$) is compared to (X), and the **reconstruction error** is minimized.

2. Learning Principle

- **Loss Function:** Mean Squared Error (MSE) between input and reconstructed output:

$$L = \frac{1}{N} \sum (X - \hat{X})^2$$

- The network learns to minimize this loss using **gradient descent** (TensorFlow) or **back-propagation** (MLPRegressor internally).
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3. Implementation Approaches

A) TensorFlow / Keras Autoencoder

- Explicitly defines encoder and decoder layers using `Dense` layers.
- Allows flexible control over architecture, activations, and latent dimensions.
- Trained using `optimizer='adam'` and `loss='mse'`.
- Provides access to latent codes and reconstructions separately.

B) Scikit-Learn Autoencoder (MLPRegressor)

- Uses a **multi-layer perceptron regressor** to map ($X \rightarrow X$).
- Hidden layers form an implicit encoder–decoder structure (e.g., 4-8-2-8-4).
- Requires minimal code and no deep-learning library.
- Optimized automatically using the Adam solver.

Both achieve the same goal — **reconstruct input data while learning its low-dimensional features**.

4. Applications

- Dimensionality reduction (alternative to PCA)
 - Noise removal (denoising autoencoders)
 - Feature extraction for classification
 - Image compression and reconstruction
 - Anomaly or outlier detection
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5. Advantages

- Learns nonlinear feature representations automatically.
- Reduces dimensionality without manual feature engineering.
- Easy to visualize 2D or 3D latent spaces.

6. Disadvantages

- Requires sufficient training data.
 - Sensitive to network architecture and learning rate.
 - May overfit if latent dimension is too large.
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Conclusion

Autoencoders demonstrate how neural networks can **self-learn compact representations** of data through reconstruction. The TensorFlow model provides full control and visualization,

while the scikit-learn version offers a simpler, lightweight implementation of the same concept. Both highlight the principle of **unsupervised feature learning** in modern AI.